

New York Solar Resource Data

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Introduction

- Data source: Data Gov API - Solar Resource data - solar data types for given US locations: <https://developer.nrel.gov/docs/solar/solar-resource-v1/>
- Output fields of interest: avg_dni, avg_lat_tilt, avg_ghi
- Goal: extract solar resource data for New York City in JSON format and convert it to a dataframe

Project Set-Up

```
library(tidyverse)
library(httr)
library(jsonlite)
```

Connect to API

```
url <- "https://developer.nrel.gov/api/solar/solar_resource/v1.json"
# parameters to retrieve New York data
parameters_list <- list(api_key = my_api, lat = 41, lon = -75)

# API requests
response <- GET(url, query = parameters_list)

# Check the status of the request
status <- status_code(response)

# Extracting content
content <- content(response, "text")
```

Parsing the JSON into an R Object

```
json_lists <- fromJSON(content) # mapping of json to R object
str(json_lists) # data structure

## List of 6
## $ version : chr "1.1.0"
## $ warnings: list()
## $ errors : list()
## $ metadata:List of 1
## ..$ sources: chr "Perez-SUNY/NREL, 2012"
```

```
## $ inputs :List of 2
## ..$ lat: chr "41"
## ..$ lon: chr "-75"
## $ outputs :List of 3
## ..$ avg_dni :List of 2
## ...$ annual : num 3.69
## ...$ monthly:List of 12
## ....$ jan: num 3.12
## ....$ feb: num 3.36
## ....$ mar: num 4.1
## ....$ apr: num 4.07
## ....$ may: num 4.15
## ....$ jun: num 4.17
## ....$ jul: num 4.6
## ....$ aug: num 4.14
## ....$ sep: num 4.02
## ....$ oct: num 3.26
## ....$ nov: num 2.58
## ....$ dec: num 2.72
## ..$ avg_ghi :List of 2
## ...$ annual : num 3.87
## ...$ monthly:List of 12
## ....$ jan: num 1.97
## ....$ feb: num 2.69
## ....$ mar: num 3.86
## ....$ apr: num 4.7
## ....$ may: num 5.45
## ....$ jun: num 5.78
## ....$ jul: num 5.98
## ....$ aug: num 5.14
## ....$ sep: num 4.23
## ....$ oct: num 2.94
## ....$ nov: num 1.99
## ....$ dec: num 1.67
## ..$ avg_lat_tilt:List of 2
## ...$ annual : num 4.52
## ...$ monthly:List of 12
## ....$ jan: num 3.55
## ....$ feb: num 4.04
## ....$ mar: num 4.86
## ....$ apr: num 4.97
## ....$ may: num 5.18
## ....$ jun: num 5.24
## ....$ jul: num 5.58
## ....$ aug: num 5.24
## ....$ sep: num 5
## ....$ oct: num 4.11
## ....$ nov: num 3.26
## ....$ dec: num 3.13
```

```
# extract output list from json_lists
```

```
outputs <- json_lists$outputs
```

```
# extract monthly vector `monthly` from avg_dni, avg_ghi and avg_lat_tilt
```

```
dni_monthly <- outputs$avg_dni$monthly
```

```

ghi_monthly <- outputs$avg_ghi$monthly
lat_tilt_monthly <- outputs$avg_lat_tilt$monthly

# create tibble from vectors, months column with month abbreviations
# on left-most column, manually adding column names
dataframe <- tibble::tibble("month" = month.abb, "avg_dni" = dni_monthly, "avg_ghi" = ghi_monthly, "avg_
#display dataframe
dataframe

## # A tibble: 12 x 4
##   month avg_dni      avg_ghi      avg_lat_tilt
##   <chr> <named list> <named list> <named list>
## 1 Jan   <dbl [1]>    <dbl [1]>    <dbl [1]>
## 2 Feb   <dbl [1]>    <dbl [1]>    <dbl [1]>
## 3 Mar   <dbl [1]>    <dbl [1]>    <dbl [1]>
## 4 Apr   <dbl [1]>    <dbl [1]>    <dbl [1]>
## 5 May   <dbl [1]>    <dbl [1]>    <dbl [1]>
## 6 Jun   <dbl [1]>    <dbl [1]>    <dbl [1]>
## 7 Jul   <dbl [1]>    <dbl [1]>    <dbl [1]>
## 8 Aug   <dbl [1]>    <dbl [1]>    <dbl [1]>
## 9 Sep   <dbl [1]>    <dbl [1]>    <dbl [1]>
## 10 Oct  <dbl [1]>    <dbl [1]>    <dbl [1]>
## 11 Nov  <dbl [1]>    <dbl [1]>    <dbl [1]>
## 12 Dec  <dbl [1]>    <dbl [1]>    <dbl [1]>

```

-> This tibble dataframe contains lists with 1 value in each.

Convert tibble to a dataframe

```

outputs_list <- json_lists$outputs
# simplify the outputs list - remove all sub-hierarchies
# from lists and return simple named vector, using unlist()
simplified_outputs_list <- unlist(outputs_list)
# restructure the vector into a 2d object (matrix())
data_matrix <- matrix(data=simplified_outputs_list, nrow=13)
# remove annual values from last row
data_matrix = data_matrix[-1,]
# convert matrix to a dataframe
df <- as.data.frame(data_matrix)
# need to append the month back, add back column names and
# put month as first column
df$month <- month.abb
names(df) <- c("avg_dni", "avg_ghi", "avg_lat_tilt", "month")
df <- df %>% select(month, everything())

```

Putting it all together

```

# combine the above steps into a single function
solar_api_json_get_df <- function(endpoint, queries = list()) {
  url <- modify_url("https://developer.nrel.gov", path = endpoint)
  response <- GET(url, query = queries)
  if (http_error(response)) {

```

```

    print(status_code(response))
    print(http_status(response))
    stop("Something went wrong.", call. = FALSE)
  }
  if (http_type(response) != "application/json") {
    stop("API did not return json", call. = FALSE)
  }
  json_text <- content(response, "text")
  json_lists <- jsonlite::fromJSON(json_text)
  outputs <- json_lists$outputs
  df <- outputs[c("avg_dni", "avg_ghi", "avg_lat_tilt")] %>%
    lapply(function(x) unlist(x$monthly)) %>%
    as.data.frame() %>%
    `rownames<-`(NULL) %>%
    mutate(month = month.abb) %>%
    select(month, everything())
  df
}

# apply the function
solar_resource_df <- solar_api_json_get_df("api/solar/solar_resource/v1.json", parameters_list)
solar_resource_df

```

```

##      month avg_dni avg_ghi avg_lat_tilt
## 1     Jan   3.12   1.97         3.55
## 2     Feb   3.36   2.69         4.04
## 3     Mar   4.10   3.86         4.86
## 4     Apr   4.07   4.70         4.97
## 5     May   4.15   5.45         5.18
## 6     Jun   4.17   5.78         5.24
## 7     Jul   4.60   5.98         5.58
## 8     Aug   4.14   5.14         5.24
## 9     Sep   4.02   4.23         5.00
## 10    Oct   3.26   2.94         4.11
## 11    Nov   2.58   1.99         3.26
## 12    Dec   2.72   1.67         3.13

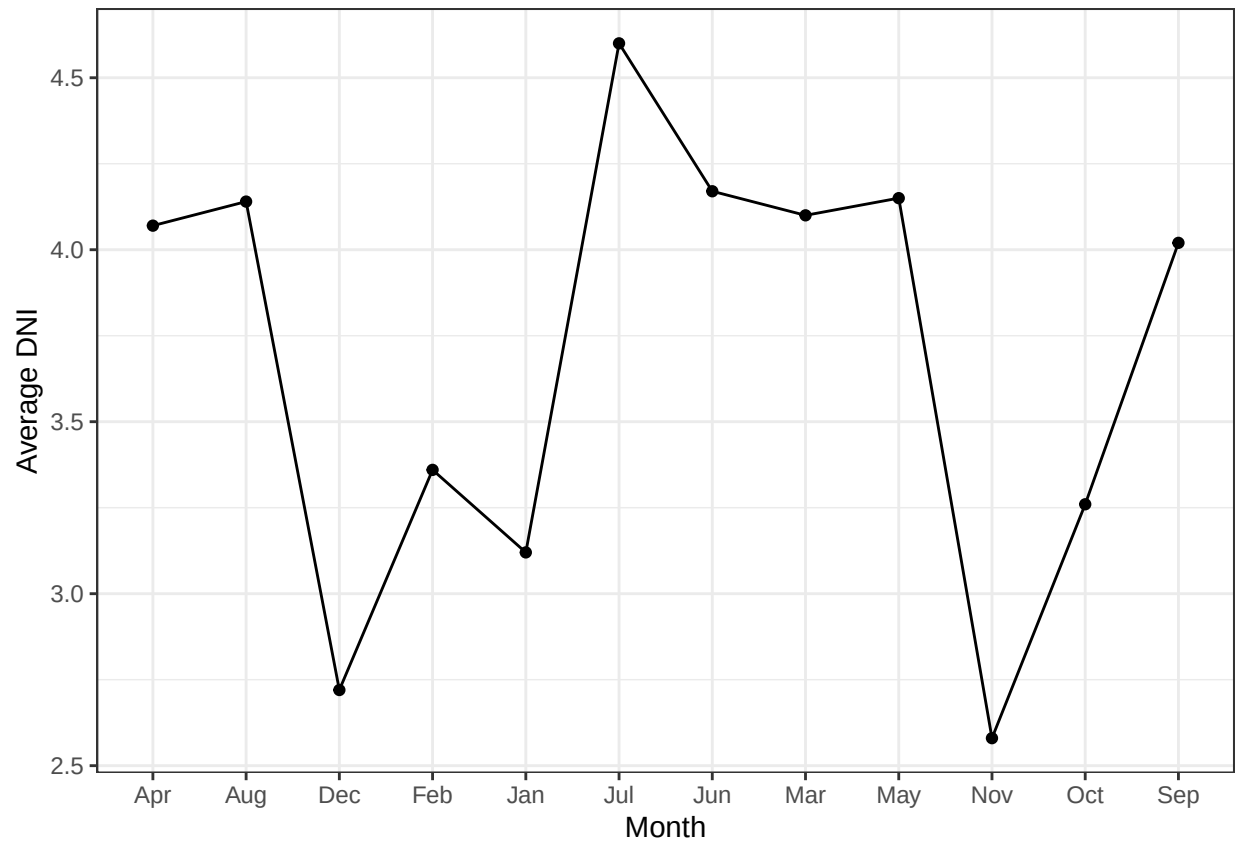
```

Visualizing New York City Solar Resource Data

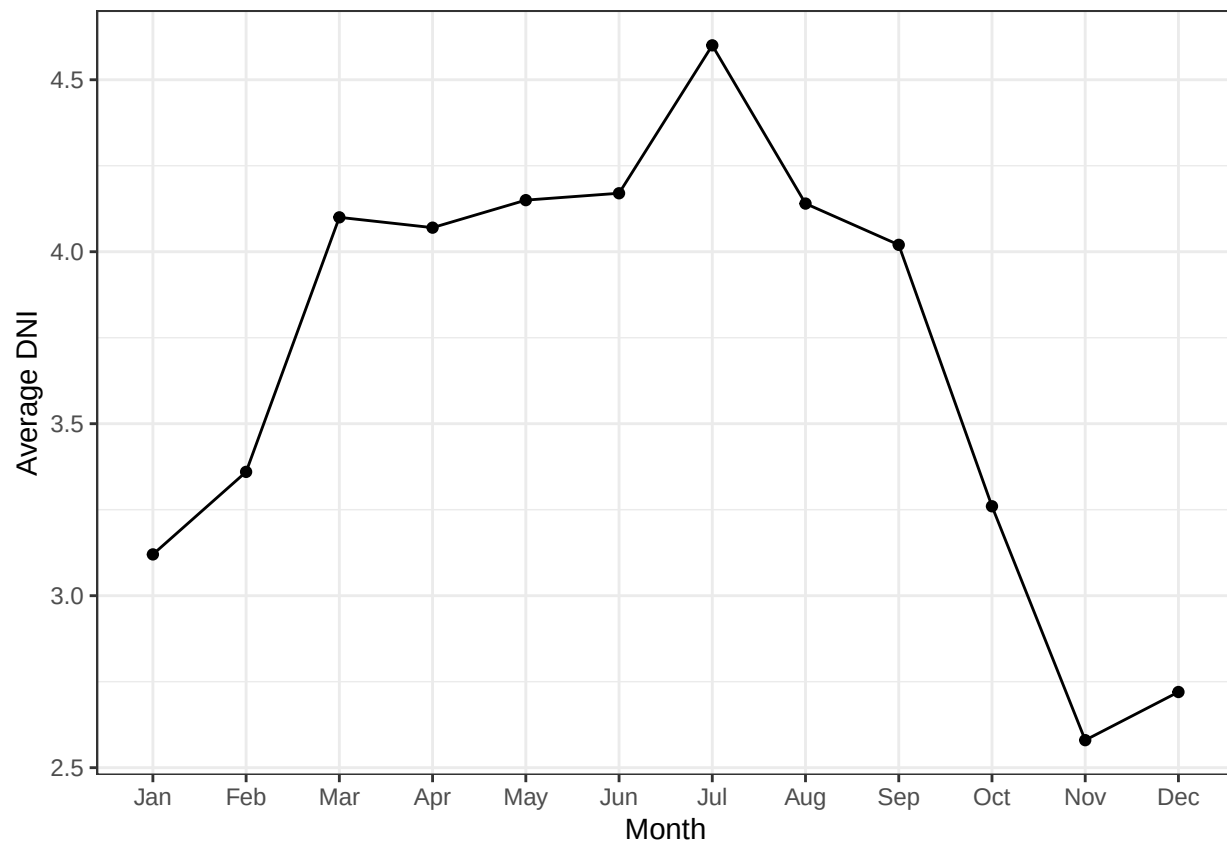
```

# x axis labels ordered alphabetically
solar_resource_df %>%
  arrange(month) %>%
  ggplot(aes(x = month, y = avg_dni)) +
  geom_line(group = 1) +
  geom_point() +
  theme_bw() + labs(x = "Month", y = "Average DNI")

```



```
# order the x axis labels by month abbreviation (Jan--> December)
solar_resource_df %>%
  mutate(month = factor(month, levels = month.abb)) %>%
  arrange(month) %>%
  ggplot(aes(x = month, y = avg_dni)) +
  geom_line(group = 1) +
  geom_point() +
  theme_bw() + labs(x="Month", y="Average DNI")
```



The operation `solar_resource_df %>% mutate(month = factor(month, levels = month.abb))` converts the month into a factor, uses the month abbreviations, and orders chronologically. Without the factor, the month abbreviations are alphabetical on the plot instead.

The figures show that, in New York, the maximum average DNI is in July (summer) and the minimum average is in November (winter), matching expectations